

KARTHIKEYAN PANDIAN PMP®, FSBI®

Project Controls & Planning Engineer

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PROFESSIONAL SUMMARY

Results-driven Project Controls and Planning Engineer with 5+ years of progressive experience in oil and gas, EPC, and offshore environments across India and Singapore. Specialises in developing logic-driven, resource-loaded Primavera P6 Integrated Master Schedules (IMS) with full Work Breakdown Structure (WBS) decomposition, critical path analysis (CPA), longest path analysis, float analysis, and resource levelling. Expert in identifying near-critical activities through longest path analysis to proactively manage schedule risk before activities become critical. Proficient in earned value management system (EVMS) reporting, delay analysis and disruption claims, look-ahead schedule preparation, risk register maintenance, and schedule compression planning. Skilled in MS Excel / Power BI dashboards, Python automation, and VBA tools delivering real-time project insights. Holds PMP® from PMI with a track record on multi-million-dollar projects for Baker Hughes (USA) and Kuraray Asia Pacific.

CORE COMPETENCIES

Schedule Management: Primavera P6 Scheduling, WBS Development, Critical Path Analysis (CPA), Longest Path Analysis, Float Analysis, Near-Critical Path Monitoring, Schedule Compression, Resource Levelling, Look-ahead Scheduling, Integrated Master Schedule (IMS), Delay Analysis, Baseline Management, Progress Measurement

Delay & Claims Analysis: Time Impact Analysis (TIA), Advanced Delay & Disruption Analysis, Windows Analysis, As-Planned vs As-Built Analysis, Collapsed As-Built, Retrospective Delay Quantification, Schedule Fragnet Development

Cost & Risk Controls: Earned Value Management (EVMS), S-Curve & KPI Reporting, Cash Flow Forecasting, Risk Register, Change Management, Stakeholder Management

Digital & Analytics Tools: Power BI, Python Automation, VBA / Excel, SAP / SAP ECTR, SQL, MS Power Platform, 4D/5D BIM – Synchro Pro

PROFESSIONAL EXPERIENCE

Project Controls / Planning Engineer | **UTOOC Engineering Pte Ltd** | Mar 2025 – Present | Singapore

Project: Kuraray ESP Project | **Client:** Wood PLC / Kuraray Asia Pacific Pte. Ltd. | **Scope:** Mechanical and Steel Structure Package – EPC

- Developed and maintained a fully logic-driven, resource-loaded Primavera P6 Integrated Master Schedule (IMS), incorporating full WBS decomposition, activity sequencing, constraint management, and critical path analysis (CPA) to ensure schedule integrity across all EPC work packages.
- Performed longest path analysis in Primavera P6 to identify the true driving path through the network, distinguishing it from the critical path to surface near-critical sequences before activities became critical and impacted the completion date.
- Conducted systematic float analysis across all schedule activities, classifying activities into critical (0 days), near-critical (1–5 days), and healthy float bands, and producing weekly float consumption trend reports to track schedule health with precision over time.
- Prepared and maintained 3-week look-ahead schedules for field execution teams, incorporating outputs from the float analysis and longest path analysis to prioritise sequencing of near-critical work fronts and prevent unplanned critical path migration.
- Produced weekly and monthly project controls deliverables including S-curves, manpower histograms, cash flow forecasts, KPI dashboards, and EVMS earned value reports — enabling senior management to make timely corrective decisions.
- Maintained the project risk register, tracking schedule risks and float consumption trends, and issued schedule compression recommendations when total float on near-critical paths fell below the agreed threshold.
- Applied Construction Measurement System (CMS) progress measurement methodology including Material Receiving Logs, Fabrication Logs, and Installation Logs to accurately quantify physical progress and calculate earned value.
- Conducted formal delay analysis and schedule impact assessments, mapping critical path deviations against baseline using Time Impact Analysis (TIA) methodology and issuing recovery plan recommendations within 48 hours of identified slippage.
- Built Power BI dashboards and Python automation scripts to replace manual Excel reporting, reducing weekly report preparation time by approximately 60% (from ~5 hours to under 2 hours per week).
- Presented daily and weekly project status reports directly to the Wood PLC client team, including critical path analysis summaries, longest path trends, and float consumption charts to maintain full schedule transparency.

Planning Engineer | **Adarsh Solutions Pvt Ltd** | Aug 2024 – Feb 2025 | Bangalore, India

Project: BH F&PT Offshore Support | **Client:** Baker Hughes, USA

Tools: Primavera P6, SAP, SQL, Python, Power BI, Synchro 4D, MS Office

- Managed the full Primavera P6 schedule lifecycle for Baker Hughes F&PT offshore support operations — including WBS development, baseline setting, critical path analysis (CPA), longest path analysis, float analysis, and look-ahead schedule preparation across a portfolio of approximately 800 activities.
- Performed weekly float analysis to classify activity criticality bands and produced float consumption trend charts, enabling the project team to redirect resources proactively to activities showing float erosion before they impacted the critical path.
- Conducted longest path analysis at each monthly schedule update to verify that the network's driving sequence aligned with the baseline critical path, and flagged any longest path migration to management as an early warning indicator of emerging delay risk.
- Maintained the project risk register and supported change management processes, assessing the schedule impact of engineering change requests via critical path analysis and issuing revised baseline recommendations with float impact quantified.
- Generated structured weekly and monthly KPI packs consolidating data from SAP and field teams via Python extraction scripts, reducing data preparation time from 3 hours to under 30 minutes.
- Performed resource levelling to resolve over-allocation conflicts on near-critical activities, balancing manpower histograms without extending the critical path.

- Created 4D construction simulations in Synchro 4D Pro to support constructability reviews and client progress visualisation sessions.

Planning Engineer | **HCL Technologies** | Mar 2021 – Aug 2024 | Coimbatore, India

Client: Baker Hughes, USA — F&PT Offshore Support (Multi-Project Engagement)

Tools: Primavera P6, MS Project, SAP, SAP ECTR, SQL, Python, Power BI, Synchro 4D, VBA

- BH F&PT Offshore Support (Mar 2021 – Aug 2024): Delivered end-to-end planning support across 3 concurrent projects with a combined schedule of over 2,000 activities — applying critical path analysis (CPA), longest path analysis, and float analysis at every monthly update cycle to maintain schedule accuracy and flag emerging risks.
- Produced monthly schedule health reports showing float distribution across the entire network, critical path analysis summaries, and longest path comparison against baseline — providing project leadership with a precise, data-driven view of schedule risk at all times.
- Database Migration — Oracle to SAP (Mar 2021 – Aug 2021): Developed and maintained the Primavera P6 project schedule, tracking milestone completion and preparing look-ahead schedules for the migration workstream.
- Spare Parts List Creation Project for BH commercial and Nuclear Valves (Mar 2022 – Dec 2023): Led scheduling and progress tracking in P6, coordinating with engineering and procurement teams to deliver a validated, client-approved spare parts register on schedule.
- Built Python and VBA automation scripts to streamline report generation, reducing weekly client status pack preparation time from 4 hours to under 45 minutes — a saving of approximately 160 hours per year.
- Developed SQL-based data extraction routines and Power BI dashboards providing live schedule, float analysis, and critical path visibility across multiple concurrent Baker Hughes work streams.

PROFESSIONAL CERTIFICATIONS

Project Management

- Project Management Professional (PMP)® — Project Management Institute (PMI) | Certification Status: **Active**
- Primavera P6 for Construction Management — CADD Centre Training Services, India | Certification Status: **Active**

Delay Analysis & Advanced Scheduling

- Expert Training in Advanced Delay and Disruption Analysis in Construction — AH Planning | Certification Status: **Active**
- Advanced Baseline Time Scheduling Techniques — EDGE PMO | Certification Status: **Active**

5D Scheduling and BIM

- SYNCHRO 4D Visual Scheduling — Bentley Systems | Certification Status: **Active**

Data and Analytics

- Full stack Power Business Intelligence FSBI® — Excel Goodies, India | Certification Status: **Active**
- MIS Reporting and Business Modeling Specialist using MS Excel — Excel Goodies, India | Certification Status: **Active**
- Data Analysis and Dashboard Associate using Power BI — Excel Goodies, India | Certification Status: **Active**
- Data Analytics and BI Developer using MS SQL and Power BI — Excel Goodies, India | Certification Status: **Active**
- Business Reporting Specialist using Python and Power BI — Excel Goodies, India | Certification Status: **Active**

Automation

- Report Automation Specialist using VBA Programming — Excel Goodies, India | Certification Status: **Active**
- VBA Developer — Excel Goodies, India | Certification Status: **Active**

Workplace Safety and Health (WSH)

- Manage Work at Height — NTUC Learning Hub, Singapore | Certification Status: **Active**
- Perform Rigger and Signalman Tasks — Green Safe, Singapore | Certification Status: **Active**
- Supervise Construction Work for WSH — NTUC Learning Hub, Singapore | Certification Status: **Active**
- Supervise Work in Confined Space Operation — NTUC Learning Hub, Singapore | Certification Status: **Active**
- Supervise Workplace Safety and Health in Process Plant — NTUC Learning Hub, Singapore | Certification Status: **Active**

EDUCATION

B.E. Mechanical Engineering — Vellalar College of Engineering and Technology (Autonomous), India | 2020 | 72% (**Verified by RMI, Singapore**)

HSC (Grade 12) — Srinivasa Vidhyalaya Matriculation School, India | 2016 | 72%

SSLC (Grade 10) — Shri Shristi Vikas Academy, India | 2014 | 83%

Industrial Training: Salem Steel Plant (Steel Authority of India) — 15 Days

*Please use the Google Drive link to access the Certificate. [link](#)